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Course: **Computational Neuroscience (BT6270)** Date: September 15, 2021

Problem 1

We have attached the MATLAB code which simulates the Hodgkin Huxley model. You need to run, and modify this code so as to find and/or plot the following:

- Threshold values for the external applied currents I_1, I_2, and I_3 in which shift of dynamical behavior from one to another is seen, such as no AP, finite number of AP's, Continuous firing and then followed by distortion resulting in no more APs.
- A graph which depicts the firing rate (frequency) as you change the applied external current (i.e. I_{ext} vs. Firing rate (f), as explained by sir in the class). You can make this plot either in MATLAB or Python.

1 Solution

1.1 Assumption

Threshold Voltage = 0 volt Channels has only two states, open and close. We also assumed that the action potential was associated with an inflow of sodium and potassium ions.

1.2 External Applied Currents

- $I_1 = 0.02238 + \epsilon$ mA, where ϵ is ± 0.00002 mA. In this range the transition happen(bifurcation), the Action potential changes to membrane potential. It goes above our threshold value (i.e. 0 V) and goes below -64 mV and again comes to its resting state potential. (See Fig. 1)
- $I_2 = 0.06234$ mA before this current value A.P was finite but beyond the application of I_2 A.P enters a continuous firing state. (See Fig. 2)
- $I_3 = 0.6283$ mA beyond this there is sudden drop in A.P spikes and only few values of membrane potential crosses the threshold. (See Fig. 3)

As we increase the current magnitude up to particular value the max value of Membrane potential(Vm) increases and as we go beyond (beyond 0.02238 mA) certain magnitude of current, Vm max value decreases but frequency of Action Potential increases but again after certain magnitude of current the Action potential frequency decreases and variation in Vm is no more similar to the Action Potential.

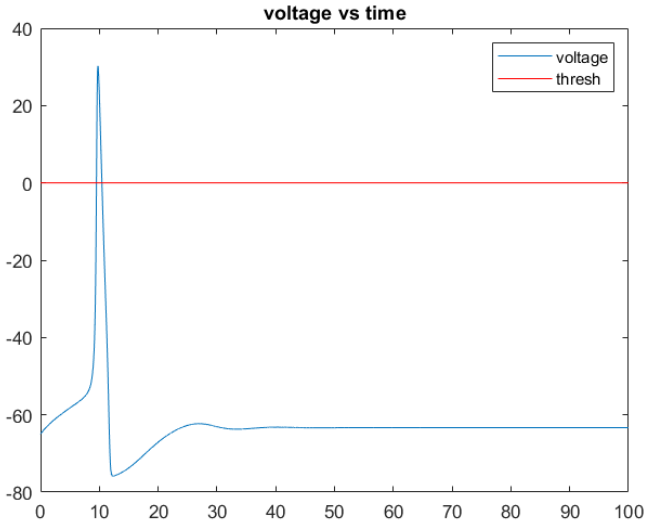

Figure 1: Voltage vs I_1

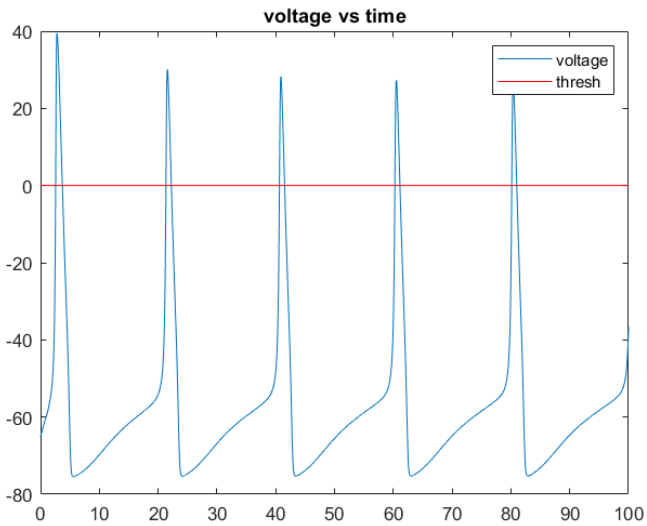

Figure 2: Voltage vs I_2

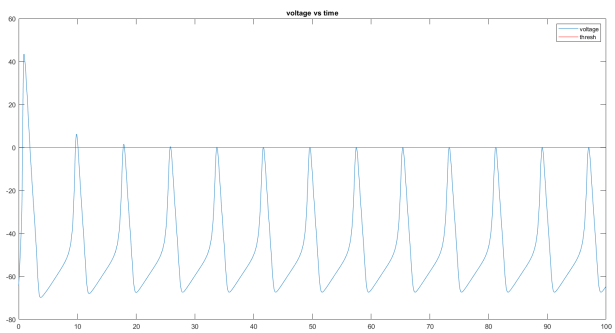

Figure 3: Voltage vs I_3

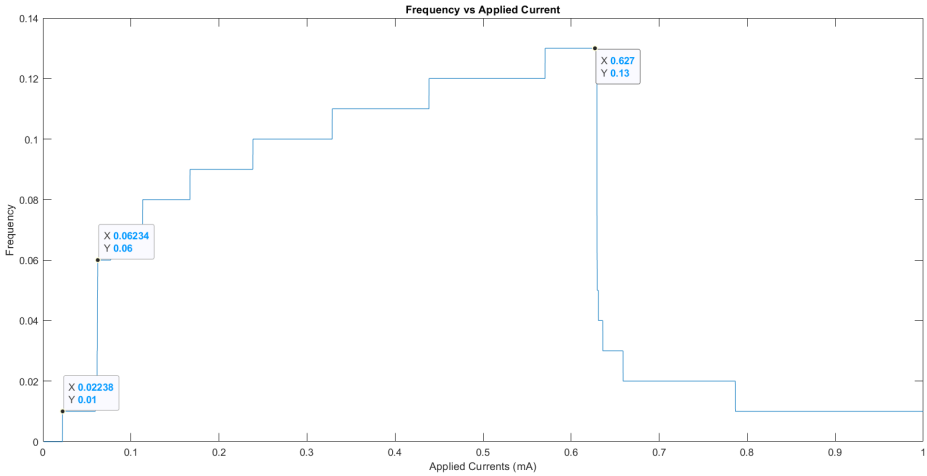

Figure 4: Current Values

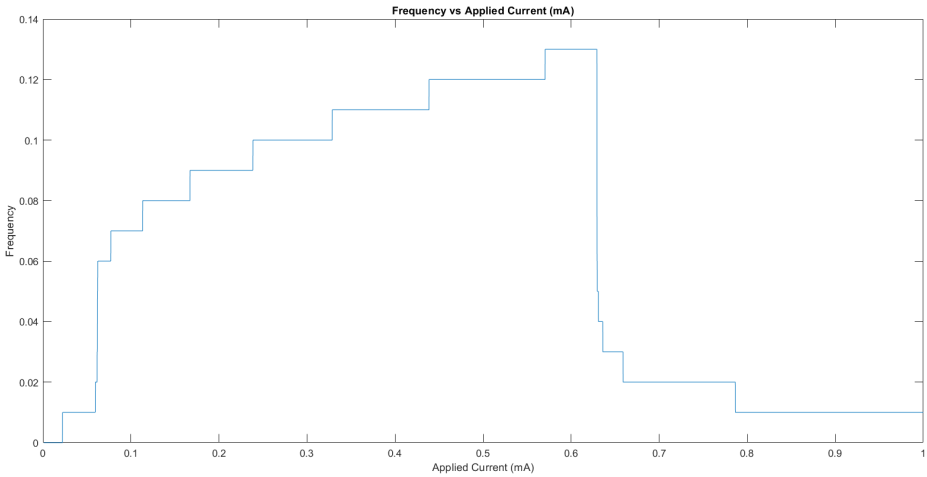

Figure 5: Voltage vs Applied Current (mA)